

The definite integral and the Newton–Leibniz formula

Two limits on the integral sign, one subtraction — and the theorem that makes area computable.

LESSON 49–51

3 × 40 MIN

ALGEBRA 11, §3.1

P1 · 10.4

LESSON CLOCK

18'

26'

26'

26'

30'

9'

Area as a limit of rectangles	18 min
The Newton–Leibniz formula	26 min
Properties	26 min
Signed area	26 min
Practice	30 min
Homework	9 min

LEARNING OBJECTIVES

- Interpret the definite integral as a limit of a sum of rectangles.
- State and apply the Newton–Leibniz formula.
- Use the properties of the definite integral.
- Explain why the constant of integration cancels.

TEXTBOOK

National	pp. 225–242
Cambridge	Pure Mathematics 1, pp. 217–226
Workbook	P1 Exercise 10D

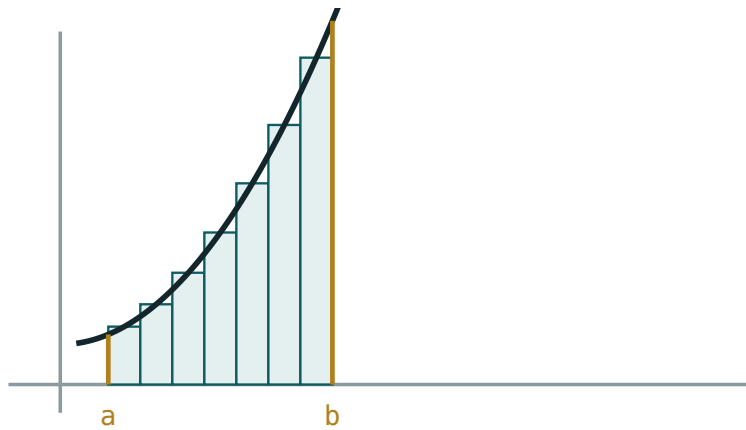
01

Explanation

Area as a limit of rectangles

To find the area under $y = f(x)$ between a and b , cut the interval into n strips, replace each strip by a rectangle, and add:

$$S_n = \sum f(x_i) \cdot \Delta x$$



more, thinner rectangles → the definite integral

Seven rectangles already approximate the area. Take more and thinner, and the error vanishes.

As $n \rightarrow \infty$ and every $\Delta x \rightarrow 0$, the sum tends to a definite number — the **definite integral**:

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum f(x_i) \Delta x$$

TWO DIFFERENT OBJECTS, ONE SYMBOL

The **indefinite** integral is a family of functions. The **definite** integral is a **number**. They are joined by the theorem below, and that link is the central result of the whole subject.

The Newton–Leibniz formula

THE FUNDAMENTAL THEOREM

If F is any antiderivative of f on $[a, b]$ then

$$\int_a^b f(x) dx = F(b) - F(a)$$

written $[F(x)]_a^b$.

Nothing about limits of sums survives in the calculation: find an antiderivative, put in the two numbers, subtract.

$$\int_0^2 x^2 dx = \left[\frac{x^3}{3} \right]_0^2 = \frac{8}{3} - 0 = \frac{8}{3}$$

NO $+C$ IN A DEFINITE INTEGRAL

The constant cancels: $(F(b) + C) - (F(a) + C) = F(b) - F(a)$. Writing it is not wrong, but carrying it through is wasted work — and writing it in the **answer** is a mistake, because the answer is a number.

Properties

$$\int_a^a f = 0 \quad \int_a^b f = -\int_b^a f$$

$$\int_a^b (f \pm g) = \int_a^b f \pm \int_a^b g$$

$$\int_a^b f = \int_a^c f + \int_c^b f$$

The last is **additivity**: an interval may be split at any interior point, and the parts add. It is what allows a curve crossing the axis to be handled piece by piece.

Signed area

THE INTEGRAL IS A SIGNED AREA

Where the curve is **below** the axis, $f(x) < 0$ and the contribution is negative. $\int_0^{2\pi} \sin x \, dx = 0$ — not because there is no area, but because the two halves cancel.

So “find the integral” and “find the area” are different questions:

1. **Integral**: compute it directly, signs and all.
2. **Area**: find the zeros of f in $[a, b]$, integrate over each piece separately, and add the **absolute values**.

For $y = x^3$ on $[-1, 1]$: the integral is 0, but the area is $\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$.

02 Worked examples

EXAMPLE 1 Evaluate $\int_1^3 (2x + 1) dx$.

① $F(x) = x^2 + x$

② $F(3) = 12, F(1) = 2.$

③ $12 - 2 = 10$

ANSWER

10

EXAMPLE 2 Evaluate $\int_1^4 \frac{1}{\sqrt{x}} dx$.

① Rewrite as $x^{-1/2}$.

② $F(x) = 2\sqrt{x}$

③ $2(2) - 2(1) = 2$

ANSWER

2

EXAMPLE 3 Find the area between $y = x^3$ and the x-axis on $[-1, 1]$.

- ① The curve crosses at $x = 0$.
Split there.

② $\int_{-1}^0 = -\frac{1}{4}$

③ $\int_0^1 = \frac{1}{4}$

- ④ Add the absolute values.

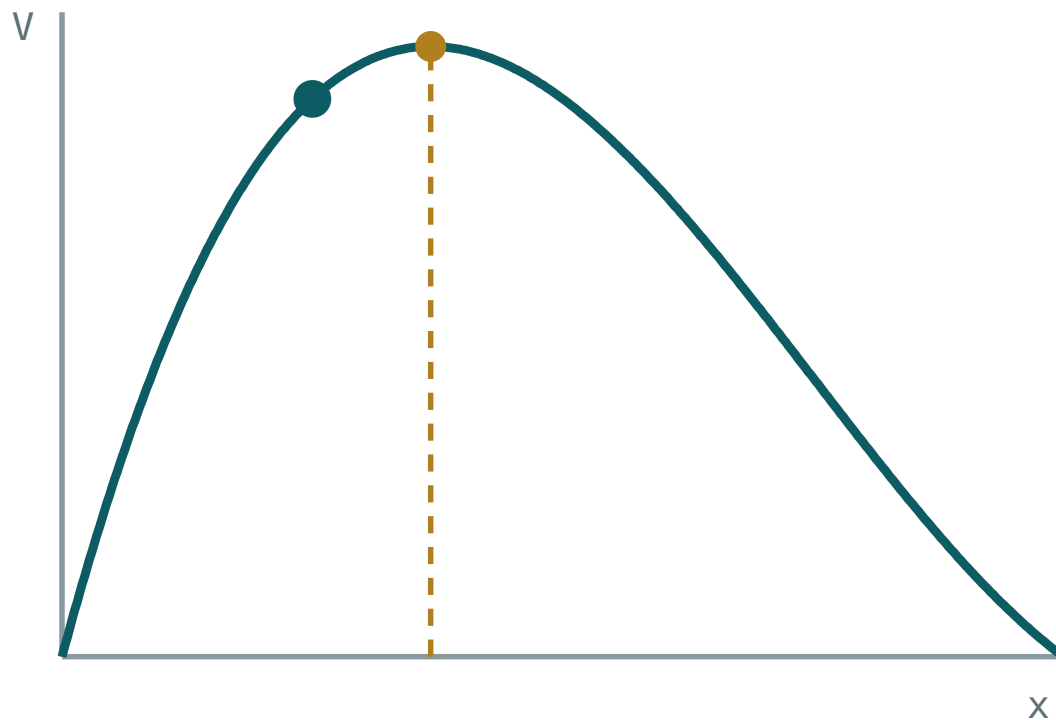
ANSWER

$$\frac{1}{2}$$

03 Interactive model

Increase the number of rectangles and watch the estimate settle on the exact value.

Rectangles filling an area



x 2

 $V(x)$ 384

greatest V 420.1

at x = 2.94

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Definite integral	Aniq integral	Определённый интеграл
Limits of integration	Integrallash chegaralari	Пределы интегрирования
Lower limit	Quyi chegara	Нижний предел
Upper limit	Yuqori chegara	Верхний предел
Newton–Leibniz formula	Nyuton–Leybnits formulasi	Формула Ньютона–Лейбница
Integral sum	Integral yig'indi	Интегральная сумма
Area under a curve	Egri chiziq ostidagi yuza	Площадь под кривой
Additivity	Additivlik	Аддитивность
Sign of the integral	Integral ishorasi	Знак интеграла

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A definite integral is:

a family of functions

a number

a derivative

an equation

The Newton–Leibniz formula gives:

$F(a) - F(b)$

$F(b) - F(a)$

$F(b) + F(a)$

$F'(b)$

The constant of integration in a definite integral:

must be included

cancels

doubles

is the answer

$\int_a^a f$ equals:

$f(a)$

0

$F(a)$

undefined

Below the axis the integral contributes:

positively

negatively

nothing

twice

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\int_0^2 x \, dx$

ANSWER 2

2. $\int_0^3 2 \, dx$

ANSWER 6

3. $\int_0^1 x^2 \, dx$

ANSWER $\frac{1}{3}$

4. $\int_1^2 3x^2 \, dx$

ANSWER 7

5. $\int_2^2 x^5 \, dx$

ANSWER 0

6. $\int_0^4 \sqrt{x} \, dx$

ANSWER $\frac{16}{3}$

7. Does a definite integral need + C?

ANSWER no

Medium · the standard question

7 PROBLEMS

1. $\int_1^3 (2x + 1) dx$

ANSWER 10

2. $\int_1^4 \frac{1}{\sqrt{x}} dx$

ANSWER 2

3. $\int_0^2 (x^2 - 3x) dx$

ANSWER $-\frac{10}{3}$

4. $\int_1^2 \frac{1}{x^2} dx$

ANSWER $\frac{1}{2}$

5. $\int_{-1}^1 x^3 dx$

ANSWER 0

6. Area between $y = x^3$ and the axis on $[-1, 1]$

ANSWER $\frac{1}{2}$

7. $\int_0^1 (3x^2 + 2x) dx$

ANSWER 2

Hard · stretch and reason

7 PROBLEMS

1. $\int_1^2 \frac{x^2 + 1}{x^2} dx$

ANSWER $\frac{3}{2}$

2. $\int_0^1 (2x + 1)^3 dx$

ANSWER 10

3. Area between $y = x^2 - 4$ and the axis on $[0, 3]$

ANSWER $\frac{23}{3}$

4. Find k with $\int_0^k x dx = 8$

ANSWER $k = 4$

5. Find k with $\int_1^k \frac{1}{x^2} dx = \frac{1}{2}$

ANSWER $k = 2$

6. Explain why $\int_0^{2\pi} \sin x dx = 0$

ANSWER The halves cancel

7. Show $\int_a^b f = \int_a^c f + \int_c^b f$

ANSWER Both equal $F(b) - F(a)$

07 Homework

Homework — 6 tasks

Write the square brackets with the limits on every answer.

1. Evaluate $\int_0^3 (x^2 + 2x) dx$.

2. Evaluate $\int_1^9 \frac{1}{\sqrt{x}} dx$.

3. Evaluate $\int_1^3 \frac{1}{x^2} dx$.
4. Find the area between $y = x^2 - 1$ and the axis on $[0, 2]$.
5. Find k with $\int_0^k 3x^2 dx = 27$.
6. Explain in three sentences why the constant of integration cancels in a definite integral.